

MUSSELS AS A SNACK OPTION TO INCREASE THE OMEGA-3 INDEX AND MARKERS OF RECOVERY IN FEMALE ATHLETES [THE MUSSELS PROJECT]

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INTRODUCTION

Long-chain omega-3 fatty acids (LC n-3s), particularly eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), support cardiovascular, neurological, inflammatory, and neuromuscular function, with emerging evidence for benefits to athlete health and recovery. (1, 2) The Australian Institute of Sport recommends achieving an Omega-3 Index (O3I) of $\geq 8\%$ as a target for athletes, using food-first approaches where practicable. (1) However, some authorities place limits on oily fish consumption, the primary dietary source of EPA and DHA, potentially limiting target intakes through food first approaches. (3) Mussels, however are not subject to the same intake limits and therefore may be an alternative food first source of EPA and DHA. (3) This study therefore will assess if supplementing the diet of active women with mussels can improve their O3I.

METHODS

A 12-week pilot, parallel-group, partially randomised controlled trial was approved by the Deakin University Human Research Ethics Committee (2025/HE000209). Healthy female endurance athletes aged 18–50 years were recruited and allocated to either a mussel or isocaloric mixed-nut control group (n=10/group), according to a Partially Randomized Patient Preference Trial (PRPT) design. (4) Participants consumed five 90-g portions of cooked mussels (450 g/week; approximately 4.5 g LC n-3/week) or 12.5 g mixed nuts, five days per week. The primary outcome was between group Omega-3 Index (O3I) change, as assessed from dried blood spots at baseline and at week 13. Dietary intake was recorded for 1 day per fortnight, while daily questionnaires delivered through REDCap monitored intervention adherence, exercise, muscle soreness, and menstrual cycle status.

EXPECTED OUTCOMES

The outcomes of The Mussels Project will contribute to determining whether mussels are a feasible food-first strategy for increasing the O3I. This may have broader relevance for female athletes, particularly where oily fish intake may be limited due to mercury exposure concerns, and where n-3 supplementation may present risks of inadvertent doping. (1, 3) Additionally, mussels may provide a more sustainable dietary source of omega-3 fatty acids compared with traditional oily fish sources. Overall, the findings of this project may help inform practical, safe, and sustainable dietary strategies for improving omega-3 status in female athlete populations. (5) Improvements in omega-3 status may also reduce perceived muscle soreness, which could support training capacity and recovery. (6) This study will provide an indication of the expected magnitude of O3I change following regular mussel consumption, which may help inform future individualised food-first strategies to support athletes in moving towards an O3I of $\sim 8\%$

RESULTS, DISCUSSION AND CONCLUSION

Data is in the process of being analysed and will be presented on the day of the symposium.

ACKNOWLEDGEMENTS

Supervisor: Lee Hamilton

Co-supervisors: Rhiannon Snipe and Lilia Convit

Research Support: Ee Ling Ng and Eloise Tarry

Technical Support: SENS Tech Team

Funding: FRDC (Fisheries Research and Development Corporation)

Mussels Supplier: Eyre Peninsula Seafoods

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